# ∞-brush: Controllable Large Image Synthesis with Diffusion Models in Infinite Dimensions

Authors: Minh-Quan Le, Alexandros Graikos, Srikar Yellapragada, Rajarsi Gupta, Joel Saltz, Dimitris Samaras

Institute: Stony Brook University

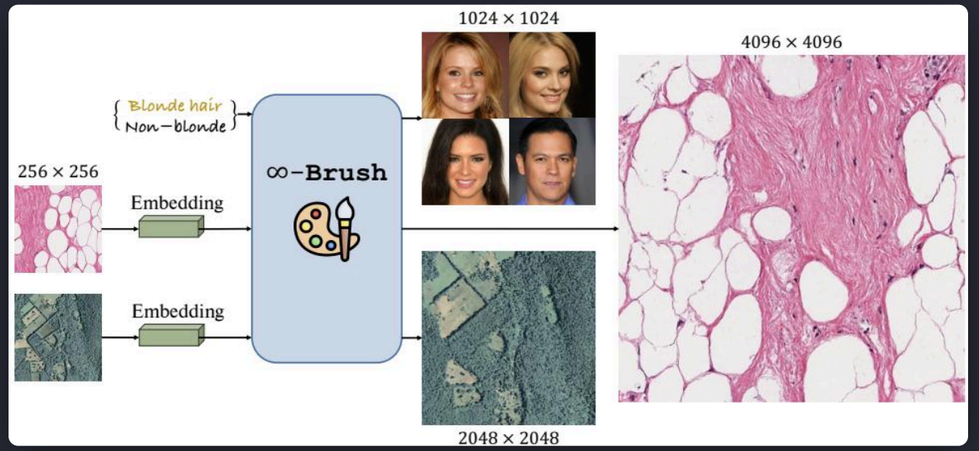
Date: August 2025

# Problem Statement

- Current diffusion models limited to fixed training resolutions
- Cannot exceed $1024 \times 1024$ without fidelity loss (e.g., SDXL)
- Patch-based methods fail to capture long-range spatial relationships
- Heavy reliance on local information during generation
- No existing model enables controllable generation at $4096 \times 4096$
- Computational demands scale quadratically with resolution

# Our Solution

- Introduce ∞-Brush: First conditional diffusion model in function space
- Generates images at arbitrary resolutions up to $4096 \times 4096$
- Key innovation: Cross-attention neural operator
- Enables conditioning in infinite-dimensional space
- Training efficiency: Only $0.4\%$ pixel subsets per image
- $65,536$ pixels instead of 16 million for $4096 \times 4096$

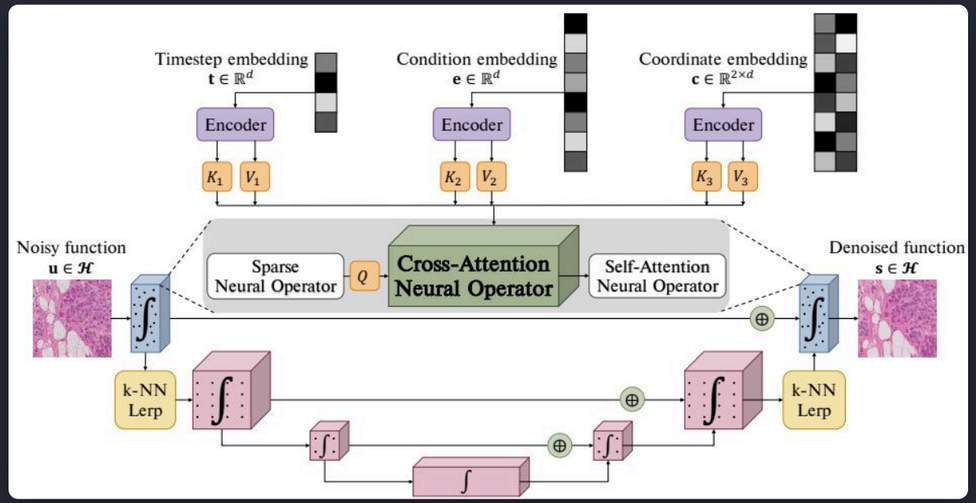


# Related Work Limitations

- **Finite-dimensional models** (SDXL, Matryoshka)
- Fixed resolution constraint: Cannot exceed training resolution
- Quadratic computational cost with resolution
- **Patch-based methods** (MultiDiffusion, Graikos et al.)
- Captures local details but loses global structure
- No conditioning for controllable generation
- **Infinite-dimensional models** ($\infty$-Diff)
- Enables arbitrary resolution but no conditional control

# Core Innovation

- **Cross-attention neural operator**
- Replaces vanilla attention in function space
- Complexity: $O((N + \Sigma N_l)d^2)$ vs. $O(N^2 d)$
- **Conditional diffusion process**
- Forward: $Q(u_t|u_0) = N(u_t; \sqrt{\bar{\alpha}_t}Au_0, (1-\bar{\alpha}_t)ACA^T)$
- **Function-space intuition**: an image is a function over continuous coordinates, not a fixed-size grid — the model learns from sparse samples yet generates at any resolution
- **vs. finite-dim & patch-based**: grid models are capped at training resolution and patches condition only locally; conditioning the whole function enables controllable arbitrary-resolution synthesis
- **Training efficiency**
- $0.4\%$ pixel subsets per image ($65,536$ vs. 16M pixels)



# Experimental Setup: Datasets & Conditioning

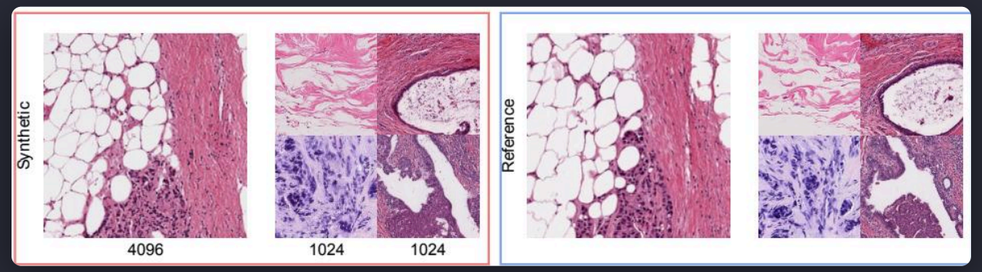
- **CelebA-HQ** · $1024 \times 1024$ · 30k faces
- Conditioning: facial attribute (blonde vs. non-blonde hair)
- **TCGA-BRCA** · $4096 \times 4096$ · 57k histopathology patches
- Conditioning: Quilt self-supervised embedding (from $256 \times 256$ downsample)
- **NAIP satellite** · $1024 \times 1024$ · 35k images
- Conditioning: DINO self-supervised embedding (ViT-B/16, $224 \times 224$)
- Training: $65,536$ pixels ($\sim 0.4\%$) per image, independent of target resolution
- Takeaway: one conditional model spans all three domains and both resolutions

# Experimental Setup: Baselines & Metrics

- **Baselines**
- SDXL (finite-dim), Graikos et al. (patch-based), ∞-Diff (unconditional)
- ∞-Diff is unconditional; ∞-Brush adds conditional control
- **Metrics**
- CLIP FID: global structure, full-resolution images
- Crop FID: local detail, $256 \times 256$ random crops
- **Evaluation resolutions**
- $1024 \times 1024$: CelebA-HQ, NAIP, BRCA 5×
- $4096 \times 4096$: BRCA 1.25×
- Takeaway: global and local quality measured separately with FID

# Results: Global Structure

- **CelebA-HQ** · 1024 × 1024 · attribute conditioning
- CLIP FID: ∞-Brush 8.38 vs. ∞-Diff 9.44
- **BRCA 5×** · 1024 × 1024 · Quilt embedding
- CLIP FID: ∞-Brush 3.74 vs. SDXL 6.64, Graikos et al. 7.43
- **NAIP** · 1024 × 1024 · DINO embedding
- CLIP FID: ∞-Brush 6.32 vs. SDXL 10.90, Graikos et al. 6.86
- **BRCA 1.25×** · 4096 × 4096 · Quilt embedding
- ∞-Brush leads Graikos et al.; SDXL cannot train at this scale
- Ablation: cross-attention operator improves CLIP FID at 4096 × 4096
- Takeaway: best CLIP FID at every tested resolution



# Results: Local Detail

- **BRCA 5×** · 1024 × 1024 · Quilt embedding
- Crop FID: ∞-Brush 17.87 vs. SDXL 6.98, Graikos et al. 15.51
- **NAIP** · 1024 × 1024 · DINO embedding
- Crop FID: ∞-Brush 48.65 vs. SDXL 11.50, Graikos et al. 43.76
- **BRCA 1.25×** · 4096 × 4096 · Quilt embedding
- Fine detail slightly behind patch-based Graikos et al.
- Reason: single global embedding vs. 16 local patch conditions
- Ablation: cross-attention operator improves Crop FID at 4096 × 4096
- Takeaway: fine detail trails both baselines at 1024 × 1024 — a modest trade-off for superior global structure

# Key Contributions

- First conditional diffusion model in function space
- Enables controllable generation at arbitrary resolutions
- Cross-attention neural operator for infinite dimensions
- Linear complexity vs. quadratic for vanilla attention
- Training on `0.4%` pixel subsets
- Achieves `4096 × 4096` synthesis with 10x lower memory

# Conclusion

- ∞-Brush overcomes resolution limits of existing models
- First to enable controllable $4096 \times 4096$ synthesis
- Key innovation: Cross-attention neural operator
- Maintains global structure while preserving details
- Computational efficiency:
- 12 hr/epoch vs. 140 hr for SDXL at $1024 \times 1024$
- Constant training cost across resolutions